

A Conditional Proof of the Strong Birch–Swinnerton-Dyer Conjecture

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Abstract

We formulate the Strong Birch–Swinnerton-Dyer formula for an elliptic curve $E/\mathbb{Q}$ as a fixed-point statement and prove a conditional theorem in that formulation. Let L^* be the leading Taylor coefficient of $L(E, s)$ at $s = 1$ and R_E the arithmetic side of the formula. We show that the Strong BSD identity holds exactly when the pair (L^*, R_E) is fixed by the averaging closure $c(a, b) = ((a + b)/2, (a + b)/2)$, and we determine what a closure on a richer layer of arithmetic data would need in order to return this statement: a readout intertwining it with c , and fixedness of the original state. The main theorem derives the identity from a recognition hypothesis, which in rank at most one contains the identity and in rank at least two is a leading-term identity with the normalization $\kappa_r = \# \text{Sha} / \# E(\mathbb{Q})_{\text{tors}}^2$. This hypothesis has the strength of the conclusion, so the theorem is a conditional translation of Strong BSD into the closure framework, not an independent derivation. The other inputs, local descent and signed hypotheses and comparison statements from the literature, are carried with their precise scope but are not used for the scalar identity. We show by examples what these inputs contain beyond coarser data: for 121a1 and 121b1 the local congruences cannot see the Tamagawa numbers 1 and 2, and for 1913b1 at $p = 3$ two generators of the same Selmer-complex determinant ideal differ in whether they have a square root. We also show that sensitivity of the BSD product to its factors does not make any hypothesis necessary, and we record a proposed overconvergent η -formula only as a conditional certificate, explaining why the published results it was modelled on do not apply as stated. The paper does not prove BSD. The scalar algebra and the closure statements are proved in Lean 4; the remaining statements are labelled by the exact extent of their formalization.

Keywords. Birch–Swinnerton-Dyer conjecture; Strong BSD; closure operator; conditional theorem; Cassels–Tate pairing; Selmer complex; Iwasawa main conjecture; formal verification (Lean); Six Birds Theory; emergence calculus.

Conventions. $E/\mathbb{Q}$ is an elliptic curve, $M_E = h^1(E)(1)$, $r = \text{ord}_{s=1} L(E, s)$ is the analytic rank, and curve labels are Cremona labels. The normalization of κ_r is taken from Tsiokos [37].

1 Introduction

For an elliptic curve $E/\mathbb{Q}$ of analytic rank r , the Strong Birch–Swinnerton-Dyer formula [4, 34] asserts

$$\frac{L^{(r)}(E, 1)}{r!} = \frac{\# \text{Sha}(E/\mathbb{Q}) \text{Reg}_{\text{NT}}(E) \Omega_E \text{Tam}(E)}{\# E(\mathbb{Q})_{\text{tors}}^2}.$$

Known results control parts of this formula: Gross–Zagier and Kolyvagin [16, 19] in rank at most one, Iwasawa main conjectures [33] for the p -parts at many primes, and Cassels–Tate and Selmer-complex methods [14, 25] for the finite arithmetic. They do not give the whole formula in general.

This paper studies the formula from the point of view of the Six Birds framework [38, 36], in which a statement is formulated as the fixed point of a *closure* (an idempotent self-map), the deep inputs are named as *recognition hypotheses*, and an audit register records how each input is accounted for. No knowledge of the framework is needed: all definitions are given here. We prove three things.

A fixed-point formulation. Write L^* for the left side and R_E for the right side of the formula, and let $c(a, b) = (\frac{a+b}{2}, \frac{a+b}{2})$ be the average of a pair and its swap. Then c is idempotent, and the Strong BSD identity holds exactly when the pair (L^*, R_E) is fixed by c (Theorem 4.4). The statement concerns the original pair: c always produces a fixed pair, and replacing an unequal pair by its average proves nothing. We also determine what a closure on a richer layer of arithmetic data would need in order to return the identity: a readout intertwining it with c , and fixedness of the original state (Theorem 4.5). We do not construct such a layer.

A conditional theorem. The recognition predicate $\Pi_{\text{BSD}}(E)$ (Definition 4.2) collects three recognition hypotheses: a local p -adic descent hypothesis $\Gamma_{\text{BSD}}^{\text{p-adic-descent}}$, a signed local hypothesis $\Gamma_{\text{BSD}}^{\text{Sha-persistence}}$ for the Sha factor, and a leading-term hypothesis $\Gamma_{\text{BSD}}^{\text{higher-GZ-fixity}}$ in rank at least two; in rank zero and one it contains the scalar identity itself. The main theorem (Theorem 9.1) states that $\Pi_{\text{BSD}}(E)$, together with comparison inputs from the literature, implies the Strong BSD identity for E .

We stress the strength of this hypothesis. In rank at least two, $\Gamma_{\text{BSD}}^{\text{higher-GZ-fixity}}$ asserts $L^* = \kappa_r \text{Reg}_{\text{NT}} \Omega_E \text{Tam}(E)$, and with the normalization $\kappa_r = \# \text{Sha} / \# E(\mathbb{Q})_{\text{tors}}^2$ of the companion paper [37] this is the identity rewritten. The scalar conclusion follows from this rank-split part of the hypothesis by explicit algebra; the local hypotheses and the comparison inputs are carried but not used. The theorem is thus a conditional translation of Strong BSD into the closure framework, with every input named, and not an independent derivation from weaker statements.

An accounting of the inputs. For each input we state exactly what is assumed and what its scope is, and we give examples showing what the inputs contain beyond coarser data. The local hypotheses are congruences modulo p -adic units, which cannot see a Tamagawa number prime to p ; for 121a1 and 121b1 at 11 the Tamagawa numbers are 1 and 2 (Section 3). The Cassels–Tate input prescribes a generator of a square-root determinant object; for the curve 1913b1 at 3, two generators of the same determinant ideal differ in whether they have a square root (Example 5.2). A proposed overconvergent η -formula is recorded only as a conditional certificate: its eigenvalue has slope 1/2 rather than the critical slope, and a cited theorem does not apply at the curves where the formula was proposed (Section 6). Finally, sensitivity of the BSD product to each factor does not make any hypothesis necessary (Remark 4.7).

Organization. Section 2 describes the comparison inputs and Section 3 the recognition hypotheses. Section 4 contains the core: the shell carrying the two sides of the formula, the recognition predicate, the fixed-point formulation and the discussion of sensitivity. Sections 5 to 7 discuss three further inputs (the Cassels–Tate comparison, the η -formula, and the p -part of the identity in rank at most one). None of the theorems of these sections is used to derive the scalar conclusion of the main theorem; the Cassels–Tate input itself remains a carried hypothesis of that theorem. Section 8 records the inputs that, to our knowledge, no current theorem supplies. Section 9 proves the main theorem, Section 10 sets up the audit register, and Section 11 lists what the paper does not claim. Many statements are checked in Lean 4 without external mathematical libraries; each footnote says

whether Lean proves the statement, a narrower version, or only records it as a conditional schema, and Appendix A gives the details.

2 Imported comparison inputs

The argument of this paper uses two kinds of hypotheses. This section describes the first kind: comparison statements from arithmetic geometry that are taken as inputs. The next section describes the second kind, the recognition hypotheses on which the main theorem actually rests. We are explicit about what each input contains, because several of them are assembled from published results whose hypotheses or normalizations differ from what the input asserts.

Each input is attached to a fixed curve: the data it compares (a map, a generator, a target, a pairing) are specified as a family indexed by the curve, and possibly a prime, before the input is asserted at the curve E of the main theorem. This prevents an input proved for one curve from being used for another.

Definition 2.1 (Cassels–Tate comparison input). Fix $E/\mathbb{Q}$ and a prime p , together with: a square-root Fitting object attached to the p -primary Cassels–Tate data of E , a target line vctp_p , a selected generator $\sqrt{F}_E^p$ of the former and a selected element $h_p^{\mathrm{CT}}(E)$ of the latter, and a property of maps between them called *canonical*. The Cassels–Tate comparison input $\chi_{\mathrm{CT},p}$ at (E, p) consists of:

1. the hypotheses that $\mathrm{Sha}(E/\mathbb{Q})[p^\infty]$ is finite, that the Cassels–Tate pairing is nondegenerate, that the Selmer-complex hypotheses of Sakamoto [31] and Macias Castillo–Sano [23] hold, and that $p \nmid \mathrm{Tam}(E)$, with p prime;
2. a map with the canonical property, sending the selected generator to the selected target element;
3. a normalized sign $\Sigma_{\mathrm{NekCT}} = +1$;
4. eight propositions $T_{E1}, \dots, T_{E8}$ recording background results from the theory of Stark systems and Selmer complexes [31, 23], determinant functors [17], and the comparison of pairings on Selmer complexes with the Cassels–Tate pairing [25, 14, 13].

The map, the image of the generator and the sign in items (2) and (3) are part of the input; they are not derived from the background results in item (4), and no statement-by-statement assignment of $T_{E1}, \dots, T_{E8}$ to published theorems is fixed here. The published results differ from the input in several specific ways. Nekovář’s comparison [25, Proposition 10.8.7] identifies the cup product on torsion classes in H^2 of a Selmer complex with *minus* the Flach pairing [14], so a sign correction and an identification of degrees are needed. The derived-height comparison of Macias Castillo–Sano [23, Theorem 4.2] assumes an odd prime of good ordinary reduction, and its relation between the Bertolini–Darmon and Nekovář pairings also carries a minus sign. Sakamoto’s theory [31] works at odd primes. The theorem of Fisher–Schaefer–Stoll [13, Theorem 6.3] normalizes the classical Cassels–Tate pairing on n -Selmer times 2-Selmer; it is not a Fitting-ideal statement and does not supply an odd-primary generator. Section 5 gives an arithmetic example showing that the determinant ideal does not determine the selected generator.

Definition 2.2 (Selmer-complex pairing input). Fix, for each curve, a target category, a Selmer complex, a pairing target and a pairing. The input $\mathcal{A}_E$ at E asserts that, for E , the Selmer complex of Nekovář [25] carries its intrinsic Cassels–Tate pairing, that the determinant-line formulation of the leading-term conjecture in the sense of Burns–Flach [6], which builds on the work of Bloch–Kato [5] and Fontaine–Perrin-Riou [15], is compatible with this Selmer complex and pairing for E (related determinant methods appear in [7, 23]); this compatibility is part of the hypothesis. It further

asserts that the pairing is bilinear (or categorical, in the categorical formulation). The notation $\mathcal{A}_E$ is local to this paper and is unrelated to the analytic leading coefficient A_E of Tsikors [37].

Definition 2.3 (Archimedean comparison input in rank at least two). Fix, for each curve, a rank, an archimedean regulator, a determinant line, an L -value side and a map from the regulator to the determinant line. The input at E asserts that the rank of E is at least 2 and that the determinant– L -value comparison of Beilinson [1] holds for E . It is required only when the analytic rank is at least 2, and its rank must equal the analytic rank of E .

These three inputs appear among the hypotheses of the main theorem (Theorem 9.1) because they describe the arithmetic setting in which its recognition hypothesis is meant to be interpreted. As we will see, the scalar conclusion of that theorem is derived from the recognition hypothesis alone; the inputs are carried alongside it and returned, not used.

Nonclaim 2.4. The comparison inputs are hypotheses. This paper does not prove them, and does not derive the map, generator image or sign of $\chi_{CT,p}$ from the cited background results. The example of Section 5 checks the hypotheses of some of those results for one curve and prime, but the prescribed generator, target, comparison map and orientation remain supplied inputs. In the formalization they are structures whose fields are the asserted statements; they introduce no axioms.

3 The recognition hypotheses

In the Six Birds framework a *recognition source* is arithmetic content that a construction rests on but does not derive: it is stated precisely, attached to the object, and used as a hypothesis. In this paper there are three, one for each of three factors of the BSD formula that the formal construction does not supply by itself.

- $\Gamma_{\text{BSD}}^{\text{padic-descent}}$ concerns the local Tamagawa factor at an odd prime of additive reduction;
- $\Gamma_{\text{BSD}}^{\text{Sha-persistence}}$ concerns the Sha factor, through a signed local comparison;
- $\Gamma_{\text{BSD}}^{\text{higher-GZ-fixity}}$ concerns the leading term in analytic rank at least 2.

Table 1 summarizes them. We say at once what will matter for the main theorem: its scalar conclusion uses only $\Gamma_{\text{BSD}}^{\text{higher-GZ-fixity}}$ in rank at least 2, and in rank at most 1 the identity itself, which the predicate contains as a hypothesis (Section 4). The two local hypotheses are part of the recognition predicate, with their scopes recorded below, but the scalar argument does not use them.

Definition 3.1 ($\Gamma_{\text{BSD}}^{\text{padic-descent}}$: p -adic descent hypothesis). Let $E/\mathbb{Q}$ and let p be an odd prime at which E has additive reduction of Kodaira type II or III, with $c_p(E) \in \{1, 2\}$, with $\text{Sha}(E/\mathbb{Q})[p^\infty] = 0$, and let $L/\mathbb{Q}_p$ be a finite tamely ramified extension over which E acquires good reduction (equivalently, over which inertia acts trivially on the ℓ -adic Tate module for every prime $\ell \neq p$; reduction of type II or III is potentially good). The hypothesis $\Gamma_{\text{BSD}}^{\text{padic-descent}}$ at (E, p, L) supplies a descent map

$$\text{descent}_p^{\text{OC}} : LC_{\text{triv}}(L_p^{\text{OC}}(E_L)) \longrightarrow LC_{\text{triv}}(L_p^{\text{OC}}(E))$$

between trivialized leading coefficients of ordinary/anticyclotomic local data over L and over $\mathbb{Q}_p$, a unit $\varepsilon_p(E, L) \in \mathbb{Z}_p^\times$, and the congruence

$$LC_{\text{triv}}(L_p^{\text{OC}}(E)) \equiv c_p(E)^{-1} \varepsilon_p(E, L) \text{descent}_p^{\text{OC}}(LC_{\text{triv}}(L_p^{\text{OC}}(E_L))) \pmod{\mathbb{Z}_p^\times}.$$

It also carries a factor tam_p , which the recognition predicate of Section 4 matches with the Tamagawa product of E .

Table 1. The three recognition hypotheses. The last column records whether the scalar argument of Theorem 9.1 uses the hypothesis.

Hypothesis	Scope	Content	Used for scalar BSD
$\Gamma_{\text{BSD}}^{\text{padic-descent}}$	odd prime p , additive type II or III, $c_p \in \{1, 2\}$, $\text{Sha}[p^\infty] = 0$, tame $L/\mathbb{Q}_p$ giving good reduction	descent congruence modulo $\mathbb{Z}_p^\times$ with factor c_p^{-1} and a unit; a local factor matched with $\text{Tam}(E)$	no
$\Gamma_{\text{BSD}}^{\text{Sha-persistence}}$	prime p of additive reduction in the signed scope, tame $L/\mathbb{Q}_p$ giving good reduction	signed corestriction determinant $\equiv c_p$ modulo $\mathbb{Z}_p^\times$; a factor matched with $\# \text{Sha}(E/\mathbb{Q})$	no
$\Gamma_{\text{BSD}}^{\text{higher-GZ-fixity}}$	no CM, analytic rank $r \geq 2$	$\psi_-(E, r) = 0$, with $\kappa_r = \# \text{Sha} / \# E(\mathbb{Q})_{\text{tors}}^2$ and matched factors; equivalent to Strong BSD for E when Sha is finite	yes, in rank ≥ 2

Two features of this hypothesis should be noted. First, since p is odd and $c_p(E) \in \{1, 2\}$, the number $c_p(E)$ is a p -adic unit (in particular $p \nmid c_p(E)$). The congruence modulo $\mathbb{Z}_p^\times$ therefore holds or fails independently of whether $c_p(E)$ is 1 or 2: it carries no information about $c_p(E)$. For example, the curves 121a1 and 121b1 have additive reduction at 11 of types II and III, with c_{11} equal to 1 and 2 respectively, and these two values have the same class modulo $\mathbb{Z}_{11}^\times$. The value of $c_p(E)$ is recorded in the hypothesis separately. Second, the matching of a local factor with the global Tamagawa product is itself a comparison. On the curve $y^2 = x^3 + 385x + 1225$, the two places 5 and 7 of type III each have $c_p = 2$, and the global Tamagawa product is 4; a single local factor is not the global one.

Definition 3.2 ($\Gamma_{\text{BSD}}^{\text{Sha-persistence}}$: signed Sha-persistence hypothesis). Let $E/\mathbb{Q}$, let p be a prime of additive reduction at which the signed Selmer theory used here applies, and let $L/\mathbb{Q}_p$ be a finite tamely ramified extension over which E acquires good reduction. Let $\text{sp}_p(E, L)$ be the signed local pairing on the cover side, $\text{cores}_{L/\mathbb{Q}_p}$ the corestriction from the cover-side target to a base-side target, and

$$C_{\text{loc}}^{\text{III}}(E, p) = \det(\text{cores}_{L/\mathbb{Q}_p} \circ \text{sp}_p(E, L))$$

the determinant of the composite. This construction, its determinant-line convention and its applicability at (E, p, L) are part of the hypothesis; the signed theory of Kobayashi and Pollack [18, 26], developed for good supersingular reduction, is background for it, not a source of it. The hypothesis $\Gamma_{\text{BSD}}^{\text{Sha-persistence}}$ at (E, p, L) asserts

$$C_{\text{loc}}^{\text{III}}(E, p) \equiv c_p(E) \pmod{\mathbb{Z}_p^\times},$$

and carries a factor sha_p , which the recognition predicate matches with $\# \text{Sha}(E/\mathbb{Q})$.

The same remark applies: when $p \nmid c_p(E)$, the congruence does not see the value of $c_p(E)$. The determinant of a composite of a map into the cover-side target with corestriction to the base side requires a determinant-line convention, which is part of the hypothesis, as is the matching of sha_p with the global order of Sha.

Definition 3.3 ($\Gamma_{\text{BSD}}^{\text{higher-GZ-fixity}}$: leading-term hypothesis in rank at least two). Let $E/\mathbb{Q}$ be without complex multiplication, of analytic rank $r \geq 2$. Put

$$\psi_-(E, r) = \frac{L^{(r)}(E, 1)}{r!} - \kappa_r(E) \text{Reg}_{\text{NT}}(E) \Omega_E \text{Tam}(E).$$

The hypothesis $\Gamma_{\text{BSD}}^{\text{higher-GZ-fixity}}$ at E asserts $\psi_-(E, r) = 0$. In the recognition predicate it is used together with the normalization $\kappa_r(E) = \# \text{Sha}(E/\mathbb{Q}) / \#E(\mathbb{Q})_{\text{tors}}^2$ and with the matching of its regulator, period and Tamagawa factors to those of E .

The name refers to the kind of theorem that would be expected to supply this hypothesis, a generalized Gross–Zagier formula in rank at least 2. No theorem supplying this normalized higher-rank identity is established among the inputs used here, and we retain it as a hypothesis (see also Remark 8.1). It is important to be clear about the strength of $\Gamma_{\text{BSD}}^{\text{higher-GZ-fixity}}$. Assume $\text{Sha}(E/\mathbb{Q})$ finite, and let Ω_E and $\text{Tam}(E) = \prod_v c_v(E)$ be as in the BSD formula. By the normalization of κ_r [37, Section 13], the arithmetic term of $\psi_-(E, r)$ is then

$$\kappa_r(E) \text{Reg}_{\text{NT}}(E) \Omega_E \text{Tam}(E) = \frac{\# \text{Sha}(E/\mathbb{Q}) \text{Reg}_{\text{NT}}(E) \Omega_E \text{Tam}(E)}{\#E(\mathbb{Q})_{\text{tors}}^2},$$

so $\Gamma_{\text{BSD}}^{\text{higher-GZ-fixity}}$, read with this normalization, is the Strong BSD identity for E . The hypothesis is not weaker than the conclusion it is used for.

Nonclaim 3.4. None of the three recognition hypotheses is derived in this paper, from the Six Birds framework or otherwise. Each is a hypothesis about a given curve, recorded in the formalization as a structure whose arithmetic content is a field, not as an axiom. We also do not claim that any of them is necessary for the scalar conclusion: in rank at least 2, $\Gamma_{\text{BSD}}^{\text{higher-GZ-fixity}}$ with its normalization already implies it.

4 The shell, its residual and its closure

This section contains the mathematical core of the paper. We first set up the object that carries the two sides of the BSD formula and the recognition hypothesis placed on it (Section 4.1). We then show that the Strong BSD identity is exactly a fixed-point statement for an explicit closure operator (Section 4.2), and explain why sensitivity of the formula to its factors does not make any hypothesis necessary (Section 4.3).

4.1 The shell and the recognition predicate

Definition 4.1 (BSD trace shell). A *BSD trace shell* $\text{Sel}_{\text{BSD}}^1$ consists of the following data.

1. A curve E and a natural number r . In the intended application E is an elliptic curve over $\mathbb{Q}$ and r its analytic rank; the definition itself treats E as a label.
2. A set k of scalars with an element 0 and operations $-$, $\cdot$ and $/$ satisfying the two laws

$$a - b = 0 \iff a = b, \quad (a/d) \cdot b = (a \cdot b)/d,$$

and elements of k standing for the leading coefficient $L^* = L^{(r)}(E, 1)/r!$, the order $\# \text{Sha}(E/\mathbb{Q})$, the regulator $\text{Reg}_{\text{NT}}(E)$, the period Ω_E , the Tamagawa product $\text{Tam}(E)$ and the squared torsion order $d = \#E(\mathbb{Q})_{\text{tors}}^2$.

3. The arithmetic side and the residual

$$R_E = \frac{\# \text{Sha}(E/\mathbb{Q}) \text{Reg}_{\text{NT}}(E) \Omega_E \text{Tam}(E)}{d}, \quad \rho_E = L^* - R_E.$$

4. Declared local scopes: a set $\mathcal{L}$ of local extensions, a *descent scope* given by two predicates, one on natural numbers n (“ n is an additive bad prime in scope”) and one on pairs (n, L) with $L \in \mathcal{L}$ (“ L is a trivializing lift at n in scope”), and a *signed scope* given by a predicate on natural numbers.
5. A set $\mathcal{F}$, the *formed layer*, with an idempotent map $C : \mathcal{F} \rightarrow \mathcal{F}$, together with data called the duality datum $J_{\text{BSD}}^!$, the visible readout $\text{Vis}_{\text{BSD}}^!$ and the audit record $\text{Audit}_{\text{BSD}}^!$.

For an actual curve one takes $k = \mathbb{R}$ with its usual operations, finite Sha, and the BSD normalizations of the period and regulator; then $\rho_E = 0$ is the Strong BSD identity. The definition keeps the operations abstract so that the formal statements can be checked on other scalar sets as well, for instance on $\mathbb{Q}$ in the control examples of Section 9. The data in item (5) are carried by the shell but play no role in the scalar argument; the map C is used only in Theorem 4.5. The formalization also attaches to the shell several propositions with supplied proofs (admissibility, lawfulness, applicability and similar bookkeeping); nothing below depends on them.

The recognition predicate collects the hypotheses of Section 3 for the curve of the shell.

Definition 4.2 (Recognition predicate Π_{BSD}). For a shell $\text{Sel}_{\text{BSD}}^!$ with curve E , the predicate $\Pi_{\text{BSD}}(E)$ consists of:

1. for every natural number n and every $L \in \mathcal{L}$ such that n and (n, L) satisfy the two predicates of the descent scope, a $\Gamma_{\text{BSD}}^{\text{padic-descent}}$ record at (E, n, L) whose factor tam_n equals $\text{Tam}(E)$;
2. for every natural number n in the signed scope, a $\Gamma_{\text{BSD}}^{\text{Sha-persistence}}$ record at (E, n) whose factor sha_n equals $\#\text{Sha}(E/\mathbb{Q})$;
3. if $r \geq 2$, a $\Gamma_{\text{BSD}}^{\text{higher-GZ-fixity}}$ record at E , of the same analytic rank, whose leading coefficient, regulator, period and Tamagawa factor are those of the shell, and whose constant is $\kappa_r(E) = \#\text{Sha}(E/\mathbb{Q})/d$;
4. if $r \in \{0, 1\}$, the identity $L^* = R_E$.

Item (4) deserves a comment. In rank 0 and 1 the predicate simply contains the scalar Strong BSD identity as a hypothesis. For many curves and primes the p -part of that identity is known, through Gross–Zagier, Kolyvagin and Iwasawa main conjectures [16, 19, 33] (Section 7), but the full identity is not known in general, and nothing in this paper establishes it. The predicate also records the local hypotheses, together with the matching of local factors with the global ones. The descent requirement is vacuous when no pair (n, L) satisfies both of its predicates, and the signed requirement is vacuous when its predicate is empty. If a requirement is not vacuous, the records it demands exist only at primes (odd primes, for descent), since every $\Gamma_{\text{BSD}}^{\text{padic-descent}}$ and $\Gamma_{\text{BSD}}^{\text{Sha-persistence}}$ record is at a prime; in particular, if the descent scope admits a pair (n, L) with n not an odd prime, or the signed scope contains an n that is not prime, then $\Pi_{\text{BSD}}(E)$ cannot hold.

Theorem 4.3 (Residual and Strong BSD). *For every shell, $\rho_E = 0$ if and only if*

$$\frac{L^{(r)}(E, 1)}{r!} = \frac{\#\text{Sha}(E/\mathbb{Q}) \text{Reg}_{\text{NT}}(E) \Omega_E \text{Tam}(E)}{\#E(\mathbb{Q})_{\text{tors}}^2}.$$

*This equivalence concerns the two scalars only. On the abstract shells of Definition 4.1 there is no general construction of $\Pi_{\text{BSD}}(E)$ from the identity.*¹

¹Lean proves the equivalence for every shell. The second statement is checked on a symbolic shell, with rational scalars and a label in place of a curve, on which the identity holds but whose declared descent scope contains the index 2, which no $\Gamma_{\text{BSD}}^{\text{padic-descent}}$ record admits. This is a statement about the abstract shell type, not an example of an elliptic curve.

Proof. By definition $\rho_E = L^* - R_E$, and by the first law of the shell $L^* - R_E = 0$ if and only if $L^* = R_E$. For the last assertion, Π_{BSD} demands records that the scalar identity does not provide; the symbolic shell described in the footnote satisfies the identity but not Π_{BSD} . $\square$

4.2 Strong BSD as a fixed-point statement

In the Six Birds framework [38] a *closure* on a set is an idempotent self-map, and an element is *closed* if the map fixes it. The Strong BSD identity has an exact formulation of this kind, with an explicit and elementary closure.

Let k be a field with $2 \neq 0$, for example $\mathbb{R}$. On $k \times k$ let $J(a, b) = (b, a)$ be the swap and let

$$c = \frac{1}{2}(\text{Id} + J), \quad c(a, b) = \left(\frac{a+b}{2}, \frac{a+b}{2} \right),$$

the average of a pair with its swap. For a shell whose scalar set is k , put $x_E = (L^*, R_E)$, the pair of the two sides of the BSD formula.

Theorem 4.4 (Strong BSD as a fixed point).

1. $c \circ c = c$; c preserves the sum $a + b$; $c \circ J = c$ and $J \circ c = c$. Moreover $c(x)$ is the only pair with equal coordinates and the same sum as x .
2. For every $x = (a, b)$: $c(x) = x$ if and only if $a = b$, if and only if $a - b = 0$.
3. For every shell whose scalar set is the field k ,

$$c(x_E) = x_E \iff L^* = R_E \iff \rho_E = 0.$$

4. If $\Pi_{\text{BSD}}(E)$ holds, then $c(x_E) = x_E$.²

Proof. (1) Both coordinates of $c(x)$ equal $(a+b)/2$, so applying c again changes nothing, the sum is $a+b$, and swapping the input or the output changes nothing. A pair (y, y) with $2y = a+b$ has $y = (a+b)/2$. (2) If $c(x) = x$ then $a = (a+b)/2 = b$; conversely if $a = b$ then $(a+b)/2 = a$. (3) is (2) applied to x_E , followed by Theorem 4.3. (4) By the scalar argument in the proof of Theorem 9.1, which uses only $\Pi_{\text{BSD}}(E)$, $\Pi_{\text{BSD}}(E)$ implies $L^* = R_E$; apply (3). $\square$

The theorem concerns the *original* pair x_E . It must not be confused with the trivial fact that $c(x)$ is always closed: for $x = (2, 1)$ one has $c(x) = (3/2, 3/2)$, which is closed, while x itself is not. Replacing an unequal pair by its average proves nothing about the pair. Equivalently, asking that c preserve the analytic coordinate, $c(x)_1 = x_1$, is already the same as asking that x be closed. Figure 1 shows the geometry, and the characteristic assumption is needed: if $2 = 0$ the average is not defined.

The closure c acts on two numbers. The Six Birds formulation of a problem asks for a closure on a richer *formed layer* $\mathcal{F}$ whose readout is the scalar pair; for BSD this would be a layer of Selmer, determinant and height data. We have not constructed such a layer. What can be proved is what such a layer would have to satisfy in order to return the scalar statement.

Theorem 4.5 (Closure on a formed layer). *Let $\mathcal{F}$ be a set with an idempotent map $C : \mathcal{F} \rightarrow \mathcal{F}$, and let $q : \mathcal{F} \rightarrow k \times k$ be a readout that intertwines C with c :*

$$q(C(x)) = c(q(x)) \quad \text{for all } x \in \mathcal{F}.$$

Write $q(x) = (q_1(x), q_2(x))$. Then for every $x \in \mathcal{F}$:

²Lean proves (1)–(4) for every field with $2 \neq 0$. In (3) the field structure need not agree with the operations of the shell: the last equivalence uses the shell's own law $a - b = 0 \iff a = b$. Statement (4) combines (3) with the scalar argument in the proof of Theorem 9.1, which uses only $\Pi_{\text{BSD}}(E)$.

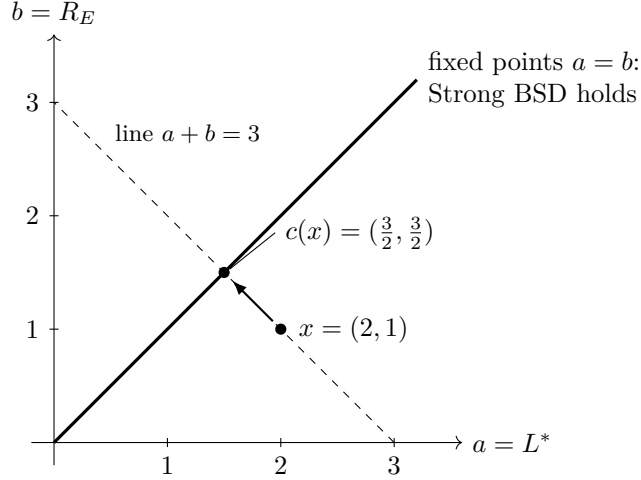


Figure 1. The averaging closure c on pairs (a, b) of an analytic and an arithmetic value. It moves each point along the line of constant $a + b$ to the diagonal. The fixed points are exactly the diagonal, where the Strong BSD identity holds. The point $x = (2, 1)$ is not fixed; its image $c(x)$ is, which says nothing about x .

1. $q(C(x)) = q(x)$ if and only if $q_1(x) = q_2(x)$;
2. if $C(x) = x$, then $q_1(x) = q_2(x)$;
3. if q is injective, then $C(x) = x$ if and only if $q_1(x) = q_2(x)$.

In particular, for a shell with formed layer $\mathcal{F}$ and closure C , an intertwining readout q and a state x with $q(x) = x_E$: if $C(x) = x$ then the Strong BSD identity holds; and if $\Pi_{\text{BSD}}(E)$ holds then $q(C(x)) = q(x)$.³

Proof. By intertwining, $q(C(x)) = c(q(x))$, so (1) is Theorem 4.4(2) applied to $q(x)$. If $C(x) = x$ then $q(C(x)) = q(x)$, and (1) gives (2). If q is injective, $q(C(x)) = q(x)$ implies $C(x) = x$, which with (2) gives (3). For the shell, (2) with $q(x) = x_E$ gives $L^* = R_E$; and $\Pi_{\text{BSD}}(E)$ gives $L^* = R_E$ by the scalar argument in the proof of Theorem 9.1, so (1) gives $q(C(x)) = q(x)$. $\square$

Three small examples over $\mathbb{Q}$ show what the injectivity, fixedness and intertwining hypotheses each contribute. Let $\mathcal{F} = (\mathbb{Q} \times \mathbb{Q}) \times \mathbb{Q}$, $C((a, b), t) = (c(a, b), 0)$ and $q((a, b), t) = (a, b)$; then C is idempotent and q intertwines. The state $((1, 1), 1)$ has equal readout but is not fixed by C , so without injectivity the converse of (2) fails. The state $((2, 1), 0)$ is not fixed, its readout is unequal, and its image $C((2, 1), 0) = ((3/2, 3/2), 0)$ is fixed: closing a state does not make the original state satisfy the identity. Finally the identity map of $\mathbb{Q} \times \mathbb{Q}$, with q the identity, fixes $(2, 1)$ but does not intertwine with c : fixedness without the intertwining identity gives nothing.

So the full closure formulation of Strong BSD rests on two inputs that this paper does not supply: a formed layer with a readout intertwining its closure with c , and the fixedness of the original state. Idempotence of a closure on its own forces neither.

4.3 Sensitivity is not necessity

The arithmetic side R_E depends on each of its factors: changing one factor while keeping the others changes R_E . This is elementary, and we record it because it is easy to over-interpret.

³Lean proves all statements, with the intertwining identity, the fixedness of the original state and the match $q(x) = x_E$ as explicit hypotheses.

Theorem 4.6 (Factor sensitivity). *Suppose the scalar set is a field, the multiplication and division of the shell are the field operations, and $\# \text{Sha}(E/\mathbb{Q})$, $\text{Reg}_{\text{NT}}(E)$, Ω_E , $\text{Tam}(E)$ and d are all nonzero. If any one of $\# \text{Sha}(E/\mathbb{Q})$, $\text{Reg}_{\text{NT}}(E)$ or $\text{Tam}(E)$ in the formula for R_E is replaced by a different value, the other factors being kept, the value of R_E changes.⁴*

Proof. In a field, multiplication by a nonzero element and division by $d \neq 0$ are injective. If, say, $\# \text{Sha}$ is replaced by $s' \neq \# \text{Sha}$, then $s' \text{Reg } \Omega \text{Tam} / d \neq \# \text{Sha} \text{Reg } \Omega \text{Tam} / d$ because $\text{Reg } \Omega \text{Tam} / d \neq 0$. The other two cases are the same. $\square$

Remark 4.7 (Sensitivity does not make a hypothesis necessary). One might hope to argue from Theorem 4.6 that each recognition hypothesis is needed, since each “governs” a factor. The argument fails. In rank at least 2, the hypothesis $\Gamma_{\text{BSD}}^{\text{higher-GZ-fixity}}$ with its normalization already implies $L^* = R_E$ (Section 3), so deleting $\Gamma_{\text{BSD}}^{\text{padic-descent}}$ and $\Gamma_{\text{BSD}}^{\text{Sha-persistence}}$ does not affect the scalar conclusion. Sensitivity of the product to its factors says nothing about which hypotheses are needed to prove an identity between the product and another number.

A genuine independence statement needs an interface in which the factors are introduced before they are normalized. Suppose native factors a and b satisfy

$$N : L^* = a \text{Reg } \Omega b, \quad C_b : b = \text{Tam}(E), \quad C_a : a = \# \text{Sha}(E/\mathbb{Q})/d.$$

Then N , C_a and C_b together imply $L^* = R_E$, and none of the three can be dropped: with $\text{Reg} = \Omega = \# \text{Sha} = d = \text{Tam} = 1$, the triples $(L^*, a, b) = (2, 1, 2)$, $(2, 2, 1)$ and $(2, 1, 1)$ satisfy exactly two of the three premises and violate the identity, since $R_E = 1 \neq 2 = L^*$. But this independence is a property of the premises, not of the identity: when $\text{Reg } \Omega \neq 0$ and N holds, $L^* = R_E$ is equivalent to the single equation $ab = (\# \text{Sha} / d) \text{Tam}(E)$, which can hold while both C_a and C_b fail (take $a = 2$, $b = 1/2$ and all other values 1). With all values positive, the solutions are exactly $a = u \# \text{Sha} / d$, $b = \text{Tam}(E)/u$ for $u > 0$. Lean checks these countermodels and the classification over $\mathbb{Q}$. None of these rational models is an elliptic curve, and we draw no arithmetic conclusion from them.

5 The Cassels–Tate comparison input

The input $\chi_{CT,p}$ of Definition 2.1 is meant to read the p -primary part of Sha through a Pfaffian, the square root of a determinant, of the Cassels–Tate pairing [9, 29]. We state what the input gives, and then show on an actual curve why its selected generator is genuinely additional data.

Theorem 5.1 (Conditional Cassels–Tate comparison). *Suppose the input $\chi_{CT,p}$ holds at (E, p) . Then p is prime, the hypotheses of item (1) of Definition 2.1 hold, and there is a map*

$$\text{Pf}_{\text{Nek},p}^{\text{or},\sqrt{\cdot}} : \sqrt{\text{Fitt}_{\text{CT}}(H^1(C_p(E))/\text{div}, U_{2,2})} \longrightarrow \text{vctp}_p$$

with the canonical property, sending the selected generator $\sqrt{F_E^p}$ to the selected element $h_p^{\text{CT}}(E)$, with normalized sign $\Sigma_{\text{NekCT}} = +1$. Here the source is the square-root Fitting object of the quotient

⁴The formalization records this statement as a conditional schema: the corresponding declaration takes a package of cancellation laws as a hypothesis, and also returns several supplied propositions (for instance, statements named computability and falsifiability) that carry no proof of what their names suggest and that we do not use. Separately, Lean proves that, when the shell’s multiplication and division are the field operations, the cancellation package holds exactly when the five displayed factors are nonzero.

by the divisible part of H^1 of the Selmer complex $C_p(E)$, for the pairing induced by Nekovář's cup product $U_{2,2}$.⁵

Proof. Each assertion is a component of the input. □

The theorem is as strong as its input and no stronger. In particular, it does not derive the map, the image or the sign from the background results listed in Definition 2.1, and the main theorem (Theorem 9.1) does not use it. Two further points, visible in the example below, concern the formulation itself. First, the published comparison of Nekovář concerns torsion classes in H^2 of a Selmer complex, with a minus sign; a source written in terms of H^1/div requires a shift of degree, which also inverts the determinant line. Second, the determinant (Fitting) ideal does not determine a selected generator, and in particular does not determine whether a prescribed generator has a square root.

Example 5.2 (An ordinary Selmer complex at $p = 3$). Let E be the curve 1913b1, $y^2 + xy = x^3 + x^2 - 34x - 135$, and $p = 3$.

1. *Arithmetic data.* E has discriminant -1913^2 , reduction of type I_2 at 1913 with $c_{1913} = 2$, rational torsion of order 2, and good ordinary reduction at 3 with $a_3 = 2$ and $\#\tilde{E}(\mathbb{F}_3) = 2$, so 3 is not anomalous and divides none of the torsion order, the Tamagawa number and the order of the reduced group $\tilde{E}(\mathbb{F}_3)$. An exact modular-symbol computation gives $|L(E, 1)|/\Omega_E = 9/2 \neq 0$; the scale of the modular symbol is fixed by the optimality of E [24, Table 1], the semistable Manin-constant theorem [11] and the structure of the real period lattice. By Gross–Zagier and Kolyvagin [16, 19], $E(\mathbb{Q})$ has rank 0 and $\text{Sha}(E/\mathbb{Q})$ is finite. The three-descent and the upper bound reported in the proof of Miller [24, Theorem 7.2] give $\text{Sha}(E/\mathbb{Q})[3^\infty] \cong (\mathbb{Z}/3\mathbb{Z})^2$.
2. *The Selmer complex.* Let C be the ordinary compact Selmer complex of $T_3(E)$ over $\mathbb{Z}_3$, with unramified conditions away from 3. The hypotheses of Macias Castillo–Sano [23] needed to describe its cohomology hold; in particular the image of Frobenius at 5, with characteristic polynomial $X^2 - 2X + 5 \equiv X^2 - 2X + 2 \pmod{3}$, generates a nonsplit Cartan subgroup of $\text{GL}_2(\mathbb{F}_3)$. One finds $H^1(C) = 0$ and $H^2(C) \cong \text{Sha}(E/\mathbb{Q})[3^\infty]^\vee \cong (\mathbb{Z}/3\mathbb{Z})^2$, and C is quasi-isomorphic to the complex $\mathbb{Z}_3^2 \xrightarrow{3J} \mathbb{Z}_3^2$ in degrees 1 and 2, with $J = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}$.
3. *The determinant ideal and two generators.* Under the canonical trivialization of the acyclic complex over $\mathbb{Q}_3$, the inverse determinant line of C has image the ideal $(9) \subset \mathbb{Z}_3$, the Fitting ideal of $H^2(C)$; the square-root ideal is (3) . The elements 9 and 18 both generate (9) , since $2 \in \mathbb{Z}_3^\times$. But $9 = 3^2$ is a square in $\mathbb{Q}_3$, while 18 is not: a square root would have valuation 1, and dividing it by 3 would give a unit whose square is 2 modulo 3, which is impossible. So every invariant that factors through the ideal, including its valuation, takes the same value on the two generators, while only one of them lies in the image of the fixed squaring map.
4. *Stark bases.* Using the theorems of Sakamoto [31] and Macias Castillo–Sano on core vertices, one obtains Stark bases over $\mathbb{Z}/27\mathbb{Z}$, and compatible bases at all levels, with identical Fitting ideals at every level and different square-root behaviour of the prescribed basis.
5. *Degree.* Since $H^1(C) = 0$ here, the quotient $H^1(C)/\text{div}$ appearing in the source of Theorem 5.1 is zero; the finite group sits in H^2 .

⁵The formalization records this as a conditional schema: the map, the image of the generator and the sign are fields of the input, and the conclusion is their projection. The data compared are fixed for the curve and prime before the input is asserted, and an input selecting a different generator or target at the same (E, p) is rejected.

The arithmetic steps in items (1), (2) and (4) are written arguments that rely on the external results cited there; the descent and the upper bound are published computations that we did not rerun. Lean checks the finite algebra: the presentation by $3J$ and its cokernel of order 9, the normalization of every perfect alternating form on $(\mathbb{Z}/3\mathbb{Z})^2$, the Cartan computation for every matrix over $\mathbb{F}_3$ with the given trace and determinant, and the square-class argument in item (3). The example does not construct the generator, target or orientation that the input $\chi_{CT,p}$ prescribes. Nor does it show that no square root can be normalized from the ideal: the generator 3 of the root ideal (3) is determined by the ideal and the trivialization. What it shows is that the square-root behaviour of a basis prescribed independently of the ideal is not determined by the ideal; the comparison of the prescribed generator, target and orientation with arithmetic objects remains part of the input.

6 The overconvergent η -formula

This section records a proposed p -adic identity at additive odd primes, the overconvergent η -formula. It plays no role in the main theorem. We state it only as a conditional input, and we explain why the published results one might hope to derive it from do not apply in the form needed.

Definition 6.1 (η -certificate). Let $E/\mathbb{Q}$ and let p be an odd prime at which E has additive reduction. An η -certificate at (E, p) consists of: a nonzero β (the proposed U_p -eigenvalue of an additive p^2 -stabilization) with $\beta^2 = -p$; a leading term $T_{\text{lead}}(E, p)$ of an overconvergent expansion, with its order $\text{ord}_T(E, p) \in \mathbb{Z}$; a value $\eta_p(E)$; nonzero scalars standing for p and $\#E(\mathbb{Q})_{\text{tors}}$; and the identity

$$\eta_p(E) = (p \#E(\mathbb{Q})_{\text{tors}})^{\text{ord}_T(E, p)-1} \beta^{-2} T_{\text{lead}}(E, p).$$

The certificate also lists, as propositions, the background results it is modelled on: Bellaïche's critical-slope theory and moment lemma [2], overconvergent modular symbols [28, 12], critical p -adic L -functions [3], the Eisenstein ideal at prime-square level [20], component groups of Néron models [22, 21], and a nonvanishing and primitivity statement for a cuspidal evaluation, which is a conjecture.

Theorem 6.2 (Conditional η -formula). *If (E, p) carries an η -certificate, then*⁶

$$\eta_p(E) = (p \#E(\mathbb{Q})_{\text{tors}})^{\text{ord}_T(E, p)-1} \beta^{-2} T_{\text{lead}}(E, p).$$

Proof. The identity is part of the certificate. □

We do not claim that the listed background results, together with the conjecture, imply the identity. Two elementary obstructions show that at least some of them do not apply as stated.

First, the relation $\beta^2 = -p$ fixes the slope of β . With the valuation normalized by $v_p(p) = 1$, it gives $2v_p(\beta) = 1$, so $v_p(\beta) = 1/2$. In weight 2 the critical slope is 1; this is the normalization of Benois–Büyükboduk [3], for whom the critical slope in weight k is $k - 1$. So β is not of critical slope, and results about critical-slope eigenvalues do not apply to it without a further comparison.

Second, the theorem of Lang–Wake [20, Theorem 1.1] concerns a level prime N and a residual prime ℓ , both at least 5, with $\ell \mid N + 1$; these numerical conditions are necessary, though not sufficient, for an application. They exclude the three curves at which the formula has been proposed, whichever residual prime is chosen. For 36a1 at $p = 3$, the level prime would be $3 < 5$ (and the full

⁶The formalization records this as a conditional schema: the identity is a field of the certificate and the conclusion is its projection.

level 36 is not prime). For 49a1 at $p = 7$, $N + 1 = 8$ has no prime divisor at least 5. For 121b1 at $p = 11$, $N + 1 = 12$ has no prime divisor at least 5.

Numerical values of η_p at these three curves have been recorded, but without complete input data or a reproducible computation of the right-hand side. We do not regard them as evidence for the formula, at those curves or in general.⁷

7 The p -part in rank at most one

In analytic rank 0 and 1, the recognition predicate contains the scalar Strong BSD identity itself as a hypothesis (Definition 4.2). This section records, for context, the form in which known results are expected to bear on that hypothesis: they give the p -part of the identity for suitable primes p . As with the other inputs, we state precisely what is assumed; the main theorem does not use this section.

Definition 7.1 (Rank-at-most-one input). The rank-at-most-one input is a list of forty propositions, each with an asserted proof, standing for published results in two branches: for curves with complex multiplication, Rubin’s Iwasawa theory over imaginary quadratic fields [30], the main conjecture at supersingular primes of Pollack–Rubin [27, Theorem 7.3], and work of Burungale–Castella–Skinner–Tian [8] at good ordinary primes, together with the archimedean, Heegner-point and uniformization ingredients needed to compare p -adic and classical quantities; for curves without complex multiplication, the work of Skinner–Urban [33], Wei Zhang [40], Castella et al. [10] and Yan–Zhu [39]. Both branches are part of the input simultaneously.

Theorem 7.2 (Conditional p -part identity in rank at most one). *Let $E/\mathbb{Q}$ have analytic rank $r \in \{0, 1\}$ and let p be a prime. Assume the hypotheses of one of the two branches:*

(CM) *E has complex multiplication by K ; either p is good ordinary and split in K , or $p \geq 5$ is good supersingular and inert in K ; $\text{Sha}(E/\mathbb{Q})[p^\infty]$ is finite, the Cassels–Tate pairing is nondegenerate, the Selmer-complex hypotheses of Sakamoto and Macias Castillo–Sano hold, $p \nmid \text{Tam}(E)$, the standard ramification and Galois-image hypotheses hold, and a Heegner point is given when $r = 1$;*

(non-CM) *E has no complex multiplication, the residual representation $\bar{\rho}_{E,p}$ is surjective [32], p is good ordinary with $p \nmid N$ and $p \nmid \text{Tam}(E)$; the local, residual-irreducibility, ramification and level hypotheses of Skinner–Urban hold; $\text{Sha}(E/\mathbb{Q})[p^\infty]$ is finite, the Cassels–Tate pairing is nondegenerate, the Selmer-complex hypotheses hold, and, when $r = 1$, the Gross–Zagier–Kolyvagin hypotheses hold for a Heegner field.*

Assume the rank-at-most-one input, and assume further that, under these branch hypotheses, the results it lists yield the valuation identity below.⁸ Then

$$v_p \left(\frac{L^{(r)}(E, 1)/r!}{\Omega_E \text{Reg}_{\text{NT}}(E)} \right) = v_p \left(\frac{\# \text{Sha}(E/\mathbb{Q}) \text{Tam}(E)}{\# E(\mathbb{Q})_{\text{tors}}^2} \right).$$

Proof. Apply the assumed implication to the branch hypotheses. □

⁷Lean proves the slope computation (from the relation $\beta^2 = -p$, the multiplicativity of the valuation at β^2 and $v_p(-p) = 1$, it derives $v_p(\beta) = 1/2 \neq 1$) and the exclusion of the level primes 3, 7 and 11 from the numerical conditions of Lang–Wake for every residual prime.

⁸The formalization records this as a conditional schema. Lean proves that the comparison hypothesis, given the forty asserted propositions and a curve and prime, is equivalent to the bare implication from the branch hypotheses to the valuation identity; the theorem is therefore exactly as strong as that implication.

The comparison hypothesis in the theorem is where the arithmetic is: in the non-CM branch, for instance, one divisibility comes from the Skinner–Urban main conjecture and the other from Euler systems and the Gross–Zagier–Kolyvagin method, and making the two match up to p -adic units requires the finiteness, nondegeneracy and nondivisibility hypotheses. We have not verified that the listed results yield the identity under exactly the stated hypotheses, and the input is not organized curve by curve. The theorem should be read as a statement of the expected form of the known p -part results, not as a new proof of them, and it does not supply the full identity required by the recognition predicate in rank at most one.

8 Inputs not supplied by known theorems

Two of the inputs above correspond to well-known open problems. We record them as remarks rather than theorems: they describe what, to our knowledge, is not currently available, and we have not carried out a systematic search of the literature to support anything stronger.

Remark 8.1 (Leading terms in rank at least two). For $E/\mathbb{Q}$ without complex multiplication and of analytic rank $r \geq 2$, we know of no theorem giving an identity of the form

$$\frac{L^{(r)}(E, 1)}{r!} = \kappa_r(E) \det(\langle Z_i, Z_j \rangle_{\text{NT}}) \Omega_E \text{Tam}(E)$$

with explicit classes $Z_1, \dots, Z_r$ forming a $\mathbb{Z}$ -basis of $E(\mathbb{Q})/E(\mathbb{Q})_{\text{tors}}$, so that the determinant is $\text{Reg}_{\text{NT}}(E)$ (for points spanning a subgroup of index m it would be $m^2 \text{Reg}_{\text{NT}}(E)$), and explicit normalizing factors, that is, a generalized Gross–Zagier formula in rank at least 2. The hypothesis $\Gamma_{\text{BSD}}^{\text{higher-GZ-fixity}}$ is exactly what such a theorem, combined with the normalization of κ_r , would supply.

Remark 8.2 (Main conjectures at additive primes 2 and 3). For $E/\mathbb{Q}$ with additive reduction at $p \in \{2, 3\}$, we know of no Iwasawa main conjecture giving an equality of characteristic ideals with explicit local correction factors. This is to be distinguished from the signed theories at primes of good supersingular reduction and from other local constructions. The hypothesis $\Gamma_{\text{BSD}}^{\text{padic-descent}}$ excludes the prime 2. The hypothesis $\Gamma_{\text{BSD}}^{\text{Sha-persistence}}$ requires separately supplied signed applicability and comparison data, and no such comparison at an additive prime 2 is established here.

Remark 8.3 (Where the local residuals are carried). In the bookkeeping of the Six Birds framework, the difficulties of the previous two remarks and of the signed local comparison appear as three residuals: a rank- r leading-term residual, a signed corestriction residual, and a local descent residual. They are not additional results. They are carried, respectively, by the hypotheses $\Gamma_{\text{BSD}}^{\text{higher-GZ-fixity}}$, $\Gamma_{\text{BSD}}^{\text{Sha-persistence}}$ and $\Gamma_{\text{BSD}}^{\text{padic-descent}}$, and naming them does not reduce what has to be supplied.

9 The conditional theorem

We can now state the main result. Figure 2 shows its logical structure: which hypotheses enter the proof of the scalar identity, and which are carried along.

Theorem 9.1 (Conditional Strong BSD). *Let $\text{Sel}_{\text{BSD}}^!$ be a shell with curve E and analytic rank r , and let C be its closure operator. Assume:*

1. *the recognition predicate $\Pi_{\text{BSD}}(E)$ (Definition 4.2);*
2. *the Cassels–Tate input $\chi_{CT,p}$ at E and some prime p , and the Selmer-complex pairing input $\mathcal{A}_E$ at E (Definitions 2.1 and 2.2);*

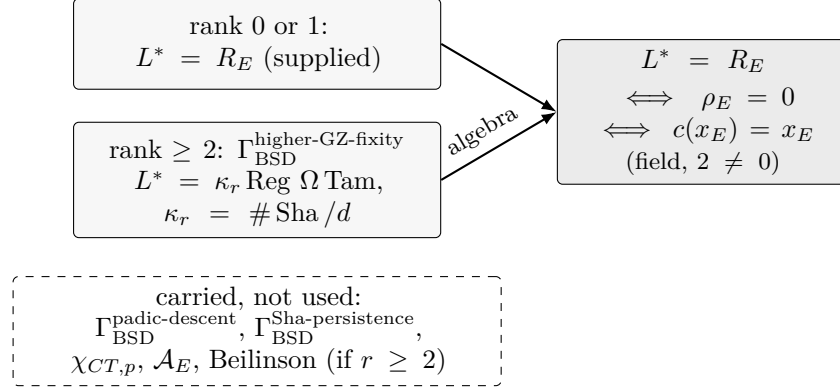


Figure 2. The logical structure of Theorem 9.1. The scalar identity is derived from the rank-split part of the recognition predicate; the equivalences on the right are Theorems 4.3 and 4.4. The dashed hypotheses are part of the theorem’s assumptions but do not enter the derivation of the identity.

3. if $r \geq 2$, the archimedean comparison input at E , of rank r (Definition 2.3).

Then

$$\frac{L^{(r)}(E, 1)}{r!} = \frac{\# \text{Sha}(E/\mathbb{Q}) \text{Reg}_{\text{NT}}(E) \Omega_E \text{Tam}(E)}{\#E(\mathbb{Q})_{\text{tors}}^2},$$

equivalently $\rho_E = 0$, and the asserted components of the inputs in (2) and (3) hold.⁹

Proof. If $r \in \{0, 1\}$, item (4) of $\Pi_{\text{BSD}}(E)$ is the identity $L^* = R_E$. If $r \geq 2$, item (3) supplies a $\Gamma_{\text{BSD}}^{\text{higher-GZ-fixity}}$ record whose factors are those of the shell and whose constant is $\kappa_r = \# \text{Sha} / d$, with d the squared torsion order. Its hypothesis $\psi_{-}(E, r) = 0$ says $L^* - \kappa_r \text{Reg } \Omega \text{Tam} = 0$, hence $L^* = \kappa_r \text{Reg } \Omega \text{Tam}$ by the first law of the shell. Substituting $\kappa_r = \# \text{Sha} / d$ and applying the second law $(a/d) \cdot b = (a \cdot b)/d$ three times gives

$$L^* = \left(\frac{\# \text{Sha}}{d} \right) \text{Reg } \Omega \text{Tam} = \frac{\# \text{Sha} \text{Reg } \Omega \text{Tam}}{d} = R_E.$$

In both cases $\rho_E = 0$ by Theorem 4.3. The remaining assertions are components of the inputs. $\square$

Three comments make the content of the theorem precise.

What the proof uses. The scalar conclusion is derived from the rank-split part of $\Pi_{\text{BSD}}(E)$ alone. The local hypotheses $\Gamma_{\text{BSD}}^{\text{padic-descent}}$ and $\Gamma_{\text{BSD}}^{\text{Sha-persistence}}$ and the comparison inputs are hypotheses of the theorem, because they belong to the arithmetic setting in which the recognition predicate is meant, but the scalar identity does not depend on them.

How strong the hypothesis is. The rank-split scalar premise contains the identity in rank 0 and 1, and expresses it through normalized fixity in rank at least 2: $\Gamma_{\text{BSD}}^{\text{higher-GZ-fixity}}$ with the normalization $\kappa_r = \# \text{Sha} / \#E(\mathbb{Q})_{\text{tors}}^2$ is the identity rewritten. The remaining hypotheses supply additional records, which the identity does not reconstruct. The theorem is therefore not an independent derivation of Strong BSD from weaker arithmetic statements, and it is not a statement about minimal hypotheses. Its content is the exact translation: the identity is equivalent to the vanishing

⁹The formalization records this theorem as a conditional schema, because it returns the components of the inputs in (2) and (3) by projection. Its scalar conclusion, however, is derived: Lean proves it from items (3) and (4) of $\Pi_{\text{BSD}}(E)$ by the algebra in the proof below. A compiled dependency check confirms that the two declarations isolating this algebra depend on no project declaration outside their own module, and that neither the derivation from $\Pi_{\text{BSD}}(E)$ nor the full theorem depends on Theorem 5.1, Theorem 6.2, Theorem 4.6, the statements of Section 10, or the supplied applicability, computability, falsifiability and ablation fields of the shell.

of ρ_E (Theorem 4.3) and, when the scalars form a field with $2 \neq 0$, to the fixedness of x_E under the averaging closure (Theorem 4.4), and it follows from the recognition predicate by explicit algebra, with every arithmetic input named and scoped.

That the hypotheses are consistent and not vacuous. The theorem has been instantiated on shells with scalars in $\mathbb{Q}$, a label in place of a curve, all factors equal to 1, and rank 0 or 2. With leading coefficient 1, all hypotheses hold and the conclusion follows; in rank 0 no archimedean input is required, and none can be supplied. With leading coefficient 2 and everything else unchanged, the recognition predicate fails, since it would force $2 = 1$. These are formal checks of the statement, not examples of elliptic curves.

Nonclaim 9.2. This is not a proof of the Birch–Swinnerton-Dyer conjecture, in its weak or strong form, for any curve. The theorem derives the Strong BSD identity for E from hypotheses whose scalar part contains it in rank at most one and expresses it through normalized fixity in rank at least two. We know of no theorem supplying the leading-term identity in rank at least two, or the main conjectures at additive primes 2 and 3 (Remarks 8.1 and 8.2).

10 The audit register

The Six Birds framework accompanies a closure argument with a register that records, for each residual difficulty, how it has been accounted for: constructed, imported from the literature, or supplied as a recognition hypothesis. Audited Operational Realisability (AOR) [36] is the framework’s notion of such an accounting; its strongest form, refinement-stable AOR, asks that the accounting survive refinement. This section sets up the register for the shell and proves a local model of refinement-stable AOR for it, in a precise and limited sense. The register is bookkeeping: it does not prove any arithmetic statement, and an entry marked as supplied remains a hypothesis.

Definition 10.1 (Audit register of the shell). A *discharge atom* consists of a residual type, a list of secondary residual types, and a status: *by construction*, *approved external input*, *bridged* (supplied as a recognition hypothesis), or one of a few other statuses, among them *zero* (eliminated). A *register* consists of an identifier, lists of observations, routes, sources, interfaces, constraints and nonclaims (the last nonempty), and a list of atoms. The register of the shell $\text{Sel}_{\text{BSD}}^1$, given recognition records $\Gamma_{\text{BSD}}^{\text{padic-descent}}$, $\Gamma_{\text{BSD}}^{\text{Sha-persistence}}$, $\Gamma_{\text{BSD}}^{\text{higher-GZ-fixity}}$ and inputs $\chi_{CT,p}$, $\mathcal{A}_E$ and the archimedean input, has eight atoms (Table 2): the construction of the shell and the normalization of κ_r , both by construction; the three comparison inputs, as approved external inputs; and the three recognition hypotheses, each bridged, with the secondary types role, target, transport and limit. Since the archimedean input requires rank at least 2, the register as set up applies to shells of analytic rank at least 2.

Theorem 10.2 (The constructed and imported entries). *In the register of the shell, the construction of the shell and the normalization of κ_r have status by construction, and the three comparison inputs have status approved external input. Moreover, if the quantities are integers or lie in a field (with the shell’s multiplication being the field multiplication) and $C = \Omega_E \text{Reg Tam}(E) \neq 0$, then one has $\kappa_r C = sC$ if and only if $\kappa_r = s$, for every s (for $s = \# \text{Sha} / \# E(\mathbb{Q})_{\text{tors}}^2$ this is the normalization of κ_r); and the asserted components of the three inputs hold.*¹⁰

Proof. The statuses are read off the register. The cancellation is Tsiokos [37, Section 13] in the integral case: if a common factor is nonzero, two multiples of it agree exactly when the multipliers agree. The last assertion restates the inputs. $\square$

¹⁰Lean proves the status assignments and the cancellation, the latter for integer-valued quantities; the components of the inputs are returned from the input records. Lean does not show that the inputs are approved in any sense beyond this label.

Table 2. The eight atoms of the audit register. “Bridged” means supplied as a recognition hypothesis; it does not mean proved.

Entry	Residual type	Status
Construction of $\text{Sel}_{\text{BSD}}^!$	presentation	by construction
Normalization of κ_r	transport	by construction
Cassels–Tate input $\chi_{CT,p}$	source	approved external input
Selmer-complex input $\mathcal{A}_E$	source	approved external input
Archimedean input (rank ≥ 2)	source	approved external input
$\Gamma_{\text{BSD}}^{\text{padic-descent}}$	source	bridged
$\Gamma_{\text{BSD}}^{\text{Sha-persistence}}$	source	bridged
$\Gamma_{\text{BSD}}^{\text{higher-GZ-fixity}}$	source	bridged

Theorem 10.3 (The recognition entries). *In the register of the shell, each of the three recognition hypotheses is an atom of residual type source, with secondary types role, target, transport and limit, and with status bridged; none has status zero. The asserted content of the three recognition records holds.*¹¹

Proof. By the definition of the register, and because the recognition records are hypotheses. $\square$

The register is *closed* if its list of nonclaims is nonempty and no atom has status zero. A *refinement* of a register keeps its identifier and all entries of its lists, and replaces each atom by atoms with the same residual type and the same status and with at least the same secondary types, every new atom refining an old one. Refinement in this sense adds detail but cannot change a status; in particular it cannot turn a bridged hypothesis into an eliminated one.

Theorem 10.4 (Refinement-stable closure of the register). *The register of the shell is closed, and every refinement of it is closed. The recognition entries satisfy Theorem 10.3, and, when $\Omega_E \text{Reg Tam}(E) \neq 0$ in the setting of Theorem 10.2, so do the constructed and imported entries.*¹²

Proof. The nonclaims list is nonempty and the eight statuses are by construction, approved external input, or bridged, none of them zero. A refinement preserves the nonclaims and the status of every atom it accounts for, so it is again closed. $\square$

Remark 10.5 (What the register does and does not say). Closedness of the register means that every residual difficulty has been given a status other than “eliminated”. Three of them are bridged: they are carried as recognition hypotheses, which this paper does not prove. The register therefore records an accounting of the conditional theorem, not a proof of Strong BSD. Its stability is proved only for the declared refinement relation, and the label “approved” on an input is not a verification of that input.

11 Scope and non-claims

The following statements delimit what the paper establishes. They are summarized in Table 3.

¹¹The formalization records this as a conditional schema: the statuses are read off the register, and the asserted content is returned from the records.

¹²Lean proves this for the refinement relation just described, which it shows to be reflexive and transitive and to preserve statuses. It does not formalize the general refinement relation of the AOR framework.

Table 3. Summary of the non-claims of Section 11.

	Boundary	Reading
1	Not a proof of BSD	Conditional translation; the scalar hypothesis has the strength of the conclusion.
2	Hypotheses not derived	Recognition hypotheses and comparison inputs are assumed.
3	No necessity	Normalized $\Gamma_{\text{BSD}}^{\text{higher-GZ-fixity}}$ alone gives the scalar identity in rank ≥ 2 .
4	No arithmetic formed layer	Intertwining readout and original fixedness are assumed.
5	No computability	Named computability fields are supplied, not proved.
6	Open inputs	Rank- ≥ 2 leading terms; additive $p \in \{2, 3\}$ main conjectures.
7	η -formula	Conditional certificate only; no evidence claimed.
8	Register	Statuses under a declared refinement; no arithmetic proved.

1. *Not a proof of BSD.* The main theorem derives the Strong BSD identity for a curve from a recognition hypothesis whose scalar part contains the identity in rank at most one and expresses it through a normalized leading-term identity in rank at least two. It is a conditional theorem and an exact translation, not an independent derivation, and it makes no claim about minimal hypotheses.
2. *The hypotheses are not derived.* The recognition hypotheses and the comparison inputs are assumed. The paper does not prove them, and does not derive the map, generator image or sign of the Cassels–Tate input from the background results it lists. The example of Section 5 checks the hypotheses of some of those results for one curve and prime, but the identification of the prescribed generator, target and orientation remains part of the input.
3. *No necessity.* We do not claim that any recognition hypothesis is necessary for the scalar conclusion. In rank at least 2, the normalized hypothesis $\Gamma_{\text{BSD}}^{\text{higher-GZ-fixity}}$ already implies it, and sensitivity of the BSD product to its factors (Theorem 4.6) is not necessity (Remark 4.7).
4. *No arithmetic formed layer is constructed.* The closure formulation (Section 4.2) returns the identity from a closure on a richer layer only under an intertwining readout and fixedness of the original state. Apart from small rational models, no such layer is constructed; for arithmetic data these are assumptions. No application of a general master theorem of the framework is claimed.
5. *No computability claims.* The formal shell carries propositions named computability and falsifiability with supplied proofs; they are not results. The scalar identity can be checked when both sides are supplied in an effective representation with decidable equality, for example over $\mathbb{Q}$, but the paper provides no procedure for computing them or for deciding the recognition predicate.
6. *Open inputs.* We know of no theorem supplying the leading-term identity in rank at least 2 or main conjectures with explicit local factors at additive primes 2 and 3 (Remarks 8.1 and 8.2). The local hypotheses are stated in restricted scopes: the descent hypothesis excludes $p = 2$, and no signed comparison at an additive prime 2 is established.
7. *The η -formula is not established.* It is recorded only as a conditional certificate; its derivation from the cited results is not claimed, two of those results do not apply as stated, and no numerical value is offered as evidence (Section 6).
8. *The register is bookkeeping.* The refinement-stable closure of the audit register (Section 10) concerns the register’s statuses under a declared refinement relation; it proves no arithmetic statement and does not formalize the general AOR theory.

12 Discussion

The classical results bearing on the Strong BSD formula each control part of it. Gross–Zagier and Kolyvagin [16, 19] relate derivatives, Heegner points, Selmer groups and regulators in rank at most one; Iwasawa main conjectures [33] give p -parts of the formula at many primes; Cassels–Tate pairings and Selmer complexes [14, 25] organize the finite arithmetic. None of them gives the whole formula in general. This paper does not change that. What it does is make one precise statement about how the formula fits a closure framework, and record exactly which inputs such a formulation needs.

The precise statement is the fixed-point formulation of Theorem 4.4: the identity holds exactly when the pair of its two sides is fixed by the averaging closure. It is elementary, and it is exact; in particular it separates fixedness of the original pair from the trivial closedness of its image. Theorem 4.5 says what a closure on a richer layer of arithmetic data would have to satisfy to return the identity: a readout intertwining it with the averaging closure, and fixedness of the original state. Neither is constructed here, and idempotence alone does not supply them.

The main theorem should be read with its hypotheses in view. Its scalar conclusion follows from a recognition hypothesis that contains the identity in rank at most one and expresses it through a normalized leading-term identity in rank at least two. The other hypotheses, the local recognition hypotheses and the comparison inputs, are carried with their scopes but do not enter the derivation. The value of the formulation therefore lies in its accounting: what is assumed, where each assumption would come from, what its scope is, and what is not known. Several points of that accounting are not obvious and are worth recording. The local congruences modulo $\mathbb{Z}_p^\times$ cannot see a Tamagawa number prime to p (Section 3). The determinant ideal of a Selmer complex does not determine whether a prescribed generator has a square root (Example 5.2). The proposed η -formula involves an eigenvalue of slope $1/2$, not the critical slope (Section 6). And sensitivity of the BSD product to its factors does not make any hypothesis necessary (Remark 4.7).

The companion paper [37] studies the same factors from the other side: which data each factor of the BSD formula retains, with finite adequacy tests and examples of information loss. Its normalization of κ_r is what identifies the rank- ≥ 2 recognition hypothesis with the Strong BSD identity here.

The same pattern, a target statement written as the fixed point of a closure with the deep inputs named as recognition hypotheses and an audit register recording their status, is the general method of the Six Birds framework [38, 36]. For BSD it yields a conditional theorem of exactly the strength of its hypotheses, together with a precise description of the inputs that an unconditional proof along these lines would require.

13 Conclusion

We placed the two sides of the Strong BSD formula in a shell and showed that the identity is equivalent to the vanishing of their difference and to the fixedness of the pair under an explicit averaging closure. We proved that the identity follows from a rank-split recognition hypothesis, which contains the identity in rank at most one and expresses it through a normalized leading-term identity in rank at least two; the theorem is therefore a conditional translation rather than an independent derivation. We stated each further input, the local recognition hypotheses and the comparisons from the literature, with its verified scope, and showed by examples where these inputs carry information that coarser data do not: the square-root behaviour of a prescribed Selmer-complex generator, and Tamagawa numbers invisible to congruences modulo units. The leading-term identity

in rank at least two, and main conjectures at additive primes 2 and 3, are inputs that, as far as we know, no current theorem supplies.

A Lean formalization and supporting data

What is formalized

The Lean 4 development accompanying this paper imports no external mathematics library. Its theorems depend only on Lean’s standard axioms (propositional extensionality, quotient soundness and classical choice, together with their standard consequences); there are no project-specific axioms, and no declaration is left unproved. The comparison inputs and the recognition hypotheses are Lean structures whose fields are the asserted statements; they are hypotheses, not axioms. None of the arithmetic geometry of the paper is formalized.

Each of the 30 labelled definitions, theorems, remarks and nonclaims falls into one of five classes, used in the footnotes and in Table 4. (The example of Section 5 is not one of these statements.)

- *Lean proves* (3 theorems): Theorems 4.3 to 4.5. The Lean statement has the content of the theorem in the body.
- *Model version* (2 theorems): Theorems 10.2 and 10.4, proved for integer quantities and for the declared refinement relation respectively.
- *Conditional schema* (6 theorems): Theorems 4.6, 5.1, 6.2, 7.2, 9.1 and 10.3. Part or all of the conclusion is obtained by projecting a supplied hypothesis. For Theorem 9.1 only the returned input components are projected: its scalar conclusion is derived, and the two declarations isolating that derivation depend on no project declaration outside their own module. A compiled dependency check confirms this, and that the derivation does not depend on the auxiliary theorems of Sections 5, 6 and 10, on Theorem 4.6, or on the supplied applicability, computability, falsifiability and ablation fields of the shell.
- *Lean definition* (11 definitions): the shell, the recognition predicate, the three recognition hypotheses, the five input records and the audit register.
- *Not formalized* (8 statements): five remarks and three nonclaims.

These classes follow the coverage records in the repository, which compare each Lean statement with the corresponding statement of the paper. The main theorem and the fixed-point formulation have controls on rational shells, including false targets that the recognition predicate rejects; rational-carrier examples test the hypotheses of the formed-layer statement (Sections 4.2 and 9).

Supporting computations

The example of Section 5 uses exact modular-symbol, local and Frobenius computations made with PARI/GP [35], the published three-descent and upper bound reported in the proof of Miller [24, Theorem 7.2], which we did not rerun, and published theorems on Selmer complexes and Stark systems [31, 23]. The local data of 121a1, 121b1 and of $y^2 = x^3 + 385x + 1225$ in Section 3 were computed with Tate’s algorithm in PARI/GP.

Data and code availability

The Lean sources, the statement-by-statement coverage records, the recorded computations and the scripts that check them are distributed with the archived release of this paper cited on the first page.

Coverage and declarations

Table 4. Lean coverage of the labelled statements in the inventory, in document order (30 rows). Labels omit the `closure:` infix.

Label	Kind	Lean coverage	Section
<code>def:chi-ct-p-import</code>	definition	Lean definition	Section 2
<code>def:a-e-import</code>	definition	Lean definition	Section 2
<code>def:beilinson-import</code>	definition	Lean definition	Section 2
<code>nonclaim:imports-not-derived</code>	nonclaim	not formalized	Section 2
<code>def:gamma-padic-descent</code>	definition	Lean definition	Section 3
<code>def:gamma-sha-persistence</code>	definition	Lean definition	Section 3
<code>def:gamma-higher-gz-fixity</code>	definition	Lean definition	Section 3
<code>nonclaim:recognition-sources-not-derived</code>	nonclaim	not formalized	Section 3
<code>def:sel-bsd-shell</code>	definition	Lean definition	Section 4
<code>def:pi-bsd</code>	definition	Lean definition	Section 4
<code>thm:pi-bsd-iff-strong-bsd</code>	theorem	Lean proves	Section 4
<code>thm:scalar-closure-translation</code>	theorem	Lean proves	Section 4
<code>thm:formed-closure-bridge</code>	theorem	Lean proves	Section 4
<code>thm:composite-signature</code>	theorem	conditional schema	Section 4
<code>rmk:sensitivity-not-necessity</code>	remark	not formalized	Section 4
<code>thm:chi-ct-p-comparison</code>	theorem	conditional schema	Section 5
<code>def:eta-published-imports</code>	definition	Lean definition	Section 6
<code>thm:oc-eta-formula</code>	theorem	conditional schema	Section 6
<code>def:t-cascade-import</code>	definition	Lean definition	Section 7
<code>thm:t-cascade-rank-le-1</code>	theorem	conditional schema	Section 7
<code>rmk:t-e12-gap</code>	remark	not formalized	Section 8
<code>rmk:t-bad-gap</code>	remark	not formalized	Section 8
<code>rmk:mode-b-residuals</code>	remark	not formalized	Section 8
<code>thm:strong-bsd-conditional</code>	theorem	conditional schema	Section 9
<code>nonclaim:landing-conditional</code>	nonclaim	not formalized	Section 9
<code>def:aor-sel-instance</code>	definition	Lean definition	Section 10
<code>thm:aor-mechanical-records</code>	theorem	model version	Section 10
<code>thm:aor-recognition-discharge</code>	theorem	conditional schema	Section 10
<code>thm:aor-instance</code>	theorem	model version	Section 10
<code>rmk:aor-partial-status</code>	remark	not formalized	Section 10

Table 5. Lean declarations for the 22 formalized statements. Labels omit the `closure:` infix; declarations omit the `SixBirdsBSD.Closure.` namespace.

Label	Lean declaration
<code>def:chi-ct-p-import</code>	<code>Imports.chiCTpImport</code>
<code>def:a-e-import</code>	<code>Imports.aEImport</code>
<code>def:beilinson-import</code>	<code>Imports.beilinsonImport</code>

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Label	Lean declaration
<code>def:gamma-padic-descent</code>	<code>RecognitionSources.gammaPadicDescent</code>
<code>def:gamma-sha-persistence</code>	<code>RecognitionSources.gammaShaPersistence</code>
<code>def:gamma-higher-gz-fixity</code>	<code>RecognitionSources.gammaHigherGZFixity</code>
<code>def:sel-bsd-shell</code>	<code>SelShell.selBSDShell</code>
<code>def:pi-bsd</code>	<code>SelShell.piBSD</code>
<code>thm:pi-bsd-iff-strong-bsd</code>	<code>SelShell.piBSDiffStrongBSD</code>
<code>thm:scalar-closure-translation</code>	<code>ScalarClosure.shellClosedIffResidualZero</code>
<code>thm:formed-closure-bridge</code>	<code>ArithmeticClosureBridge.formedClosureForcesStrongBSD</code>
<code>thm:composite-signature</code>	<code>SelShell.compositeSignature</code>
<code>thm:chi-ct-p-comparison</code>	<code>ChiCTp.chiCTpComparison</code>
<code>def:eta-published-imports</code>	<code>EtaFormula.etaPublishedImports</code>
<code>thm:oc-eta-formula</code>	<code>EtaFormula.ocEtaFormula</code>
<code>def:t-cascade-import</code>	<code>TCascade.tCascadeImport</code>
<code>thm:t-cascade-rank-le-1</code>	<code>TCascade.tCascadeRankLeOne</code>
<code>thm:strong-bsd-conditional</code>	<code>Landing.strongBSDConditional</code>
<code>def:aor-sel-instance</code>	<code>AORInstance.aorSelInstance</code>
<code>thm:aor-mechanical-records</code>	<code>AORInstance.aorMechanicalRecords</code>
<code>thm:aor-recognition-discharge</code>	<code>AORInstance.aorRecognitionDischarge</code>
<code>thm:aor-instance</code>	<code>AORInstance.aorInstance</code>

B Version history

v1, 31 August 2026

First public version.

v2, 4 October 2026

Revised after a statement-by-statement audit of the mathematics and of its Lean formalization. Changes:

- The main theorem is stated with the strength of its hypothesis made explicit: its scalar conclusion is derived from the rank-split part of the recognition predicate, and the other hypotheses are carried but not used. The claims that a closure master theorem applies, that each recognition hypothesis is indispensable, and that the recognition predicate is computable and falsifiable are withdrawn; the factor-sensitivity statement is retained with a remark explaining why it does not imply necessity.
- New results: the fixed-point formulation of Strong BSD under the averaging closure, and the conditions under which a closure on a richer layer returns it.
- The comparison inputs are attached to the curve of the theorem, with the archimedean input required only in rank at least two; the Cassels–Tate input states its map, generator, target and sign as hypotheses, with the discrepancies from the cited literature described and an example on 1913b1 at $p = 3$ added.
- The local recognition hypotheses state their scopes; it is noted that their congruences modulo units do not see Tamagawa numbers prime to p , and that matching local and global factors is a further comparison.

- The η -formula is recorded only as a conditional certificate; its claimed derivation from published results and its numerical anchors are withdrawn, and two scope obstructions are given.
- The rank-at-most-one statement makes its comparison hypothesis explicit; the two literature no-go theorems become remarks; the audit register is described as a local model of refinement-stable AOR.
- Lean coverage is described statement by statement; the text is rewritten throughout for readability.

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